Calculus test-bank system

My role Template and item design

I built a parent-item template for consistent variants, then drafted calculus questions with solution and distractor notes.

ORIGINAL DOCUMENT EXCERPT

SME Parent Item Draft Template

Project: Early Transcendental Test Bank

Brief SME instruction: Complete one copy of this template for each parent item. Use the dropdowns where provided. Delete optional sections that do not apply before submitting the item for review. Keep math notation in the format requested by the project lead.

1. Item Metadata

2. Stimulus / Question Stem

Include all information the student needs, including any directions, context, data, equations, graph references, or table references.

3. Answer Content

Delete what isn't needed.

Multiple Choice / Multiple Select Answer Options

For non-MC items, delete this table. Keep each rationale aligned to the same error pattern across variants.

Fill-in-the-Blank / Numeric Entry / Free Response Answer

For MC or MS items, delete this table if it is not needed.

4. Worked Solution